

Large-Scale Offshore Agile Tailoring: Exploring Product and Service Organisations

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ABSTRACT

This paper uses empirical data to compare large-scale agile method tailoring in two types of global software development programme: offshore outsourced services and offshore captive development centres. This research forms part of a larger study and uses data collected from face-to-face semi-structured open-ended interviews with over 50 practitioners in 9 international companies selected using a snowball sampling technique. Analysis has been conducted using ‘Glaserian’ grounded theory concepts, including open coding, memoing, constant comparison and theoretical saturation.

The contribution of the paper is to present a novel classification model contrasting solution focused development, where offshore organisations deliver a complete solution, and execution focused development, where offshore organisations implement work packages defined by the client. The research has discovered evidence of solution and execution focused models used by outsourced service providers. In addition, evidence is presented of solution and execution focused models used by insourced offshore development centres.

The classification model contrasts offshore project management with offshore management disintermediation, where staff from offshore organisations are directly managed by onshore client staff. Product ownership is commonly located onshore, but here too, counter examples of offshore product ownership have been found in outsourced service providers and insourced offshore development centres.

CCS Concepts

•Social and professional topics → Offshoring; •Software and its engineering → Agile software development; *Programming teams*;

Keywords

Large-scale, agile, off-shoring, outsourcing, far-shoring, insourcing, in-house development centres, sourcing models

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XP '16 Workshops, May 24 2016, Edinburgh, Scotland UK

© 2016 ACM. ISBN 978-1-4503-4134-9/16/05...\$15.00

DOI: <http://dx.doi.org/10.1145/2962695.2962703>

1. INTRODUCTION

This paper contrasts tailored models of agile software development used in large-scale outsourced offshore software development programmes and in-house offshore development centres. A combination of technological innovation and the evolution of work, business processes and national policies is driving globalization in software development [2]. Outsourcing describes the process of procuring products or services from a third-party vendor or provider [15]. Outsourcing can be performed within a particular territory, onshore outsourcing; or by using a geographically remote third party vendor, offshore outsourcing. Further, large-enterprises establish in-house offshore development centres. This paper focuses on far-shoring where onshore and offshore centres are separated by significant geographical, temporal and cultural distances.

In this study, large-scale development programmes consist of more than 25 developers configured into multiple cooperating teams, working together for a period of 9 months or more. A development programme comprises a portfolio of inter-operating software products or services whose delivery to customers is coordinated.

The research question for this study is “how do practitioners describe contrasting models of far-shore large-scale software development programmes” This research question was addressed through a grounded theory analysis of open-ended semi-structured interviews with over 50 practitioners from 9 international companies.

2. LARGE-SCALE OFFSHORE AGILE

Agile methods comprise a range of responsive development processes that comprise incremental development using feature-based multi-disciplinary teams [7]. Agile methods have tended to be associated with enhanced face-to-face communication for smaller co-located teams. There is evidence that agile methods improve both software development productivity and product quality [8].

2.1 Large-Scale Agile

Scaling, portfolio management, inter-team coordination and architecture are important concerns in using agile methods for large-scale development programmes [6]. Large enterprises using agile methods have been adopting scrum practices [3]. Large-scale development programmes are hampered by knowledge boundaries that must be overcome by development teams by using various practices, roles and artefacts [14]. The artefacts used in large-scale agile projects include those more commonly associated with plan-based or waterfall projects, such as risk assessments, reference archi-

tections and release plans [5]. Further, product ownership tends to be conducted by teams comprising product sponsors, domain specialists and programme architects [4].

2.2 Offshoring and Outsourcing

Globalised models of software development have become standard in large-scale development programmes. Global sourcing and cross-site collaboration adds a set of specific risks, in addition to the regular development programme management challenges. In-house offshore development centre decision-making processes have been investigated [16]. The three main challenges are communication complexity, coordination and control over physical, temporal and cultural distances [11]. Issues of cultural distance have been investigated [12]. There is evidence that medium-sized companies have difficulty establishing successful offshore outsourcing relationships [13].

This research suggests a potential extension to the taxonomy developed by Šmite et al. [17] by establishing management configurations used by in-sourced and outsourced teams in far offshore projects with large temporal distances.

3. METHODS

This research adopts a qualitative, grounded theory approach to investigate process tailoring in large-scale software development programmes [10, 1]. This paper uses semi-structured interviews with over 50 practitioners collected from within nine international companies to investigate software engineering practice.

3.1 Research Sites

Large enterprises engaged in off-shore outsourced or in-house offshore software development programmes were selected for this study, as shown in Table 1. The companies were identified using a snowball sampling techniques drawing on a network of professional contacts from previous employment in the commercial software sector.

3.2 Data Collection

Three sources of evidence were obtained in this study: documents produced by the companies in the study, observations of workplace practices and face-to-face semi-structured interviews with practitioners.

Practitioner interviews were recorded, transcribed, carefully reviewed and coded. The interviews were based on a semi-structured interview guide and were open-ended to provide respondents with the opportunity to raise any issues or concerns that were outside the scope of the scripted interview questions. Probing questions were used to explore interesting topics raised by interviewees and to garner depth of study data.

3.3 Data Analysis

A ‘Glaserian’ grounded theory approach was employed for data analysis [9]. Open coding, in which each code was represented by a short descriptive phrase, was used. The codes were subsequently classified into concepts to organise the large volume of data into categories. Memo writing, in the form of short, often informal, essays on a topic including selected quotations, was used to capture and sharpen the topics identified. Constant comparison, in which the researcher iterates back and forth between data collection and analysis was used to compare incidents that apply to each category,

integrate categories and their properties. Over the course of the study, theoretical saturation occurred when new research sites or interviews no longer result in new categories in the area of research under investigation.

4. FINDINGS

In this study, all the outsourcing enterprises, whether outsourced or in-house, located most front-line programming staff off-shore, “all development teams are solely based in India” (Director of Engineering, Company E). This is presumably motivated by the perceived benefits of a lower cost base offshore.

However, large enterprises with in-house offshore development centres have a variety of design and programming job functions in different locations,

“we’ll have part of engineering, part of QA teams, UI teams and the user design teams in one location and part of the team in another location. In some other projects we’ll have some functions only in one location and other functions in another location. So we’ll have all kinds of mixes” (Product Manager, Company B).

This implies a globally distributed approach to agile software development.

4.1 Product and Project Management

The geographical location of product ownership varies,

“the [core framework] product owners are based in California, for [credit card] based in New Zealand, for [marketing campaign management] the product owner is based in India” (Director of Engineering, Company E).

Hence, sometimes product ownership is offshore, but most commonly it is onshore. This is because product ownership is closely aligned to the application business domain and domain knowledge is most commonly with the client onshore. Even in in-house offshore development centres, such as Company F, the offshore expertise is in software development, while the application domain expertise, such as in healthcare, is retained onshore.

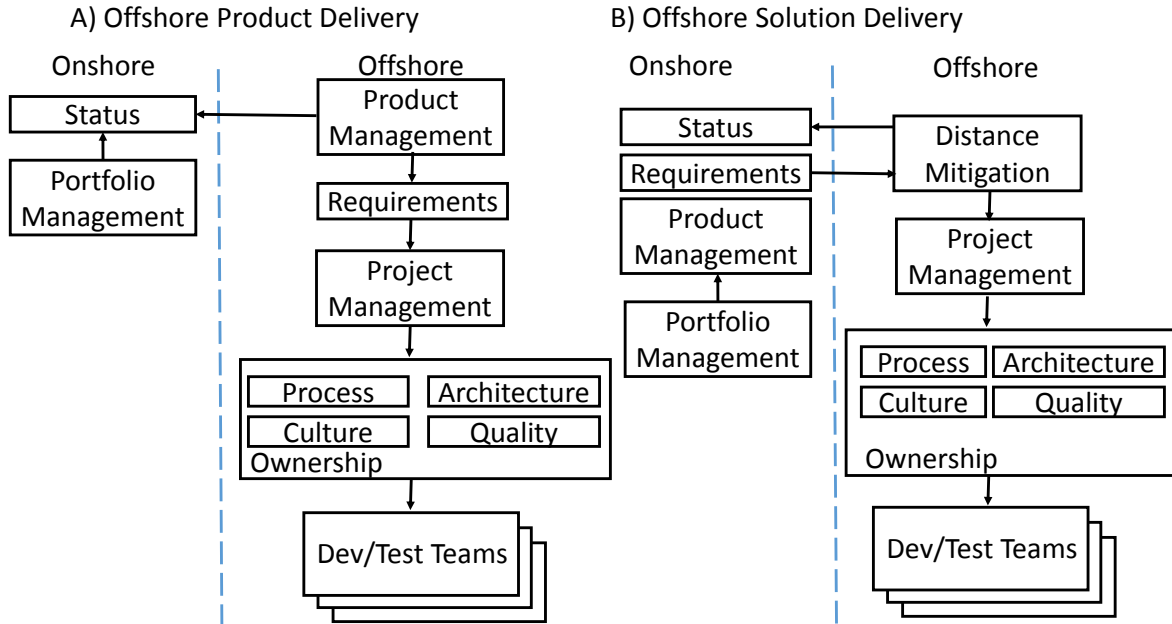
Although in this study product ownership is most commonly onshore, project management of the development teams, in contrast, is most commonly offshore. However, this study also found evidence of onshore project management such as in extended workbench arrangements,

“this project is like a staff augmentation project... The project has something like 32 staff... [the client] knows the requirements, [the offshore staff] are interacting with their business. Michelle is a [client Scrum Master] technical expert who already knows about their application. She has five staff members, in India, working for her” (Project manager, Company H).

In this case, all levels of project management, even including team leaders, were onshore, only the development team members were offshore. The daily stand-up meeting are conducted by the onshore scrum master, using conferencing technology with the offshore team.

Table 1: Participating Companies and Team Details

Company	Company Sector	Project	Team Size	Process
A	IT Service Provider	CRM (Insurance)	325	RUP
		CRM (Banking)	50	Scrum
		CRM (Healthcare)	75	Scrum
B	Internet	Internet (Calendar)	25	Scrum
		Internet (Mail)	25	Scrum
		Internet (Options)	25	Scrum
C	Software Service Provider	Transport (Rail Ticketing)	40	XP
D/E	Software Service Provider	Marketing (Campaign Management)	25	Scrum
		Enterprise CRM (Core)	20	Scrum
		Enterprise CRM (Banking)	12	Scrum
		Enterprise CRM (Credit Card)	20	FDD
		Enterprise CRM (Financial Services)	25	RUP
F	Industrial Products	Healthcare	1000	Scrum
		Energy Management	300	Lean
		Building Technologies	250	Scrum
G	IT Service Provider	Media Entertainment	50	Scrum
		Healthcare	180	Scrum
H	IT Service Provider	Travel (Loyalty)	30	Scrum
		Travel (Airline Reservation)	25	Scrum
		Risk Management and Insurance	30	Scrum
I	Enterprise CRM	CRM	300	Scrum

**Figure 1: Offshore Solution Models.**

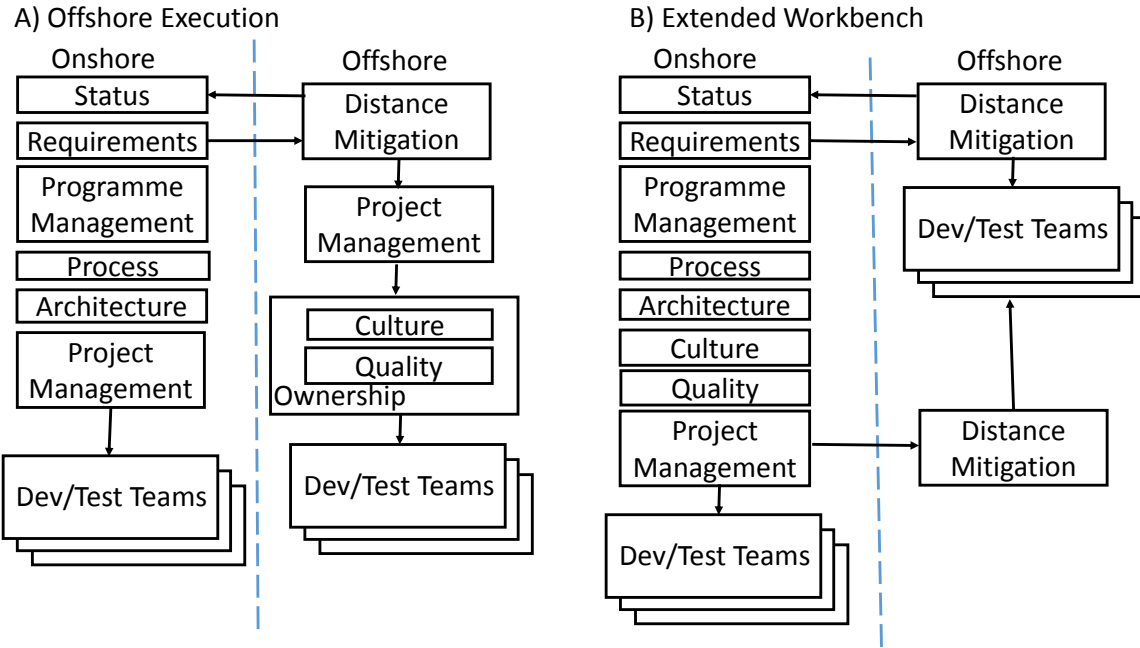


Figure 2: Offshore Execution Models.

4.2 Development Process Ownership

One offshore outsourcing service provider identifies three types of onshore client adopting agile methods. First,

“an existing customer, we have been working for over a period of time, decides to try out Agile” (Chief Technology Officer, Company H).

In this case, the outsourcing service provider may have an opportunity to influence the client’s selection of development method, but the final decision on development method resides with the customer. Second,

“a [new] customer already practising Agile engaging with us for the first time” (Chief Technology Officer, Company H).

In this case, the outsourcing service provider must adapt to the client’s chosen development method and any tailoring the client has implemented. And finally, third,

“a new customer that is also is new [to agile], then we have the leeway of recommending a method” (Chief Technology Officer, Company H).

In this third case, the outsourcing service provider is promoting their chosen development method, based on their knowledge, experience and expertise in the field. Hence, outsourcing service providers only select development methods when clients have little software development expertise.

4.3 Architecture Ownership

For offshore outsourcing service providers, clients can dictate architecture choices,

“the client wanted to move to a newer technology stack, they wanted a service [oriented] architecture” (Development Manager, Company C)

and

“[the client] has their own integral architecture team; which is governing [decision-making]” (Corporate Lead Architect, Company H).

However, offshore outsourcing service providers routinely develop their own awareness of architecture best practices

“we have a [vendor]-wide architecture council and quality council” (Scrum Master, Company D).

In-house offshore development centres does not always own their own architectures,

“the [onshore] engineering team tells the offshore [development] team ... these are the engineering tasks we want you to perform” (Product Manager, Company B).

These findings show that architectural design, bearing in mind large enterprises are often using a software product line model, like product ownership tends to be located onshore close to application domain expertise.

5. DISCUSSION & CONCLUSIONS

Four categories of offshore software development independent of the in-house/outsource procurement model have been discovered in this study.

5.1 Offshore Product Delivery

In offshore product delivery, product ownership is offshore co-located with software development, as shown in Figure 1, A. Portfolio management and product sponsorship maybe onshore, but the offshore development centre can establish its own product management infrastructure, product architecture and the opportunity to develop an indigenous development process. Captive offshore development centres

have clients and programme sponsors from within their organisation's line management or from other business units within the enterprise. Professional relationships can be built over successive projects and longer term development programmes. These longer term relationships can benefit from open and transparent communication which produces enhanced levels of trust.

5.2 Offshore Solution Delivery

In an offshore solution delivery approach, product ownership is retained onshore but decisions about the development process and product architecture are made in the offshore development centre, as shown in Figure 1, B. The offshore development centre is responsible for end-to-end delivery of the overall solution. Large-scale software development may not be the client's core business and the offshore team will have an established track record of delivering programmes with their delivery model and a given product architecture style.

5.3 Offshore Execution

In an execution oriented approach, onshore decisions determine development processes and product architectures as shown in Figure 2, A. This is typically because the onshore client wishes to retain consistency with on-shore development teams.

From an offshore service provider perspective, each client has different working cultures and different ideas about development processes and software architecture which must be managed.

In this configuration, the outsourcing vendor surrounds the team with local, knowledgeable and accessible domain experts and project management support. It is these domain experts that conduct an intensive knowledge transfer during (sometimes prolonged) visits to onshore clients, in order to then be able to return offshore and support development team members.

5.4 Offshore Extended Workbench

In an extended workbench model, even project management is conducted onshore as shown in Figure 2, B. Hence, product architecture, development process and product ownership and project management, sometimes even including team leadership are onshore. In some cases onshore scrum masters have been found managing offshore team members. In this staff augmentation model, the outsourcing vendor merely provides staff members to join teams run by the client.

Outsourcing vendors must excel at using diverse development processes and product architectures that maybe imposed by different clients. Customers may retain control over the software processes and product architectures by monitoring vendor compliance and mandating changes. Customer relationships are important for all software development organisations but for outsourcing vendors client relationships are arbitrated, in the last resort, by contracts.

6. REFERENCES

- [1] S. Adolph, W. Hall, and P. Kruchten. Using grounded theory to study the experience of software development. *Empirical Software Engineering*, 16(4):487–513, 2011.
- [2] W. Aspray, F. Mayadas, and M. Y. Vardi. Globalization and Offshoring of Software: A Report of the ACM Job Migration Task Force, 2006.
- [3] J. M. Bass. Influences on Agile Practice Tailoring in Enterprise Software Development. In *Agile India*, pages 1 – 9, Bangalore, India, Feb. 2012. IEEE.
- [4] J. M. Bass. How product owner teams scale agile methods to large distributed enterprises. *Empirical Software Engineering*, 20(6):1525 – 1557, 2015.
- [5] J. M. Bass. Artefacts and agile method tailoring in large-scale offshore software development programmes. *Information and Software Technology*, 74:1 – 16, Jul 2016.
- [6] T. Dingsøyr and N. Moe. Towards principles of large-scale agile development. In T. Dingsøyr, N. Moe, R. Tonelli, S. Counsell, C. Gencel, and K. Petersen, editors, *Agile Methods. Large-Scale Development, Refactoring, Testing, and Estimation*, volume 199 of *Lecture Notes in Business Information Processing*, pages 1–8. Springer International Publishing, 2014.
- [7] T. Dingsøyr, S. Nerur, V. Balijepally, and N. B. Moe. A decade of agile methodologies: Towards explaining agile software development. *Journal of Systems and Software*, 85(6):1213 – 1221, 2012. Special Issue: Agile Development.
- [8] T. Dybå and T. Dingsøyr. What do we know about agile software development? *IEEE Software*, 26(5):6–9, 2009.
- [9] B. G. Glaser. *Basics of Grounded Theory Analysis: Emergence vs. Forcing*. Sociology Press, Mill Valley, USA, 1992.
- [10] B. G. Glaser and A. L. Strauss. *Discovery of Grounded Theory: Strategies for Qualitative Research*. Aldine, Chicago, IL., USA, 1967.
- [11] G. Höfner and V. Mani. TAPER: A generic framework for establishing an offshore development center. In *Global Software Engineering, 2007. ICGSE 2007. Second IEEE International Conference on*, pages 162–172, Aug. 2007.
- [12] S. Krishna, S. Sahay, and G. Walsham. Managing Cross-cultural Issues in Global Software Outsourcing. *Commun. ACM*, 47(4):62–66, Apr. 2004.
- [13] N. B. Moe, D. Šmite, G. K. Hanssen, and H. Barney. From offshore outsourcing to insourcing and partnerships: four failed outsourcing attempts. *Empirical Software Engineering*, 19(5):1225–1258, Oct. 2014.
- [14] K. H. Rolland. Scaling Across Knowledge Boundaries: A Case Study Of A Large-Scale Agile Software Development Project. Edinburgh, UK, June 2016.
- [15] S. Sahay, B. Nicholson, and S. Krishna. *Global IT Outsourcing: Software Development across Borders*. Cambridge University Press, Nov. 2003.
- [16] D. Šmite, C. Wohlin, A. Aurum, R. Jabangwe, and E. Numminen. Offshore insourcing in software development: Structuring the decision-making process. *Journal of Systems and Software*, 86(4):1054–1067, Apr. 2013.
- [17] D. Šmite, C. Wohlin, Z. Galviņa, and R. Prikladnicki. An empirically based terminology and taxonomy for global software engineering. *Empirical Software Engineering*, 19(1):105–153, July 2012.